Follicular lymphoma

Indolent B-cell lymphoma — **advanced disease is not curable; survival is usually many years**. Treat by symptoms and burden (GELF); **POD24 and transformation** are the events that change prognosis. Old grade 3B is now **follicular large B-cell lymphoma — treat as DLBCL**

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin — follicular lymphoma · NCCN · primary trial reports verified against PubMed, September 2026.

BIOLOGY AND CLASSIFICATION

t(14;18) IGH::BCL2 *~85% of classic FL*	BCL2 juxtaposed to the IGH enhancer → **BCL2 overexpression blocks apoptosis**. Germinal-center phenotype: **CD20+, CD10+, BCL6+, BCL2+; CD5–, CD23–** (usually). CREBBP / KMT2D / EZH2 mutations are common secondary hits
Classic FL (cFL) *WHO-HAEM5*	Old grades 1, 2 and 3A. **Grading is now optional** — poorly reproducible and of little value for individual treatment. **Grade 3A is managed as FL**
Follicular large B-cell lymphoma	WHO-HAEM5 name for **old grade 3B** (sheets of centroblasts, no centrocytes; often CD10– and t(14;18)–). **Treat as DLBCL** with R-CHOP. ICC 2022 keeps the name "FL grade 3B"
Variants that should not be over-treated	**Pediatric-type FL** (young, localized, BCL2–, excellent outcome) · **duodenal-type FL** (low dissemination risk) · **in situ follicular B-cell neoplasm** (incidental)

WORKUP

- **Excisional or generous core biopsy** — architecture and centroblast content cannot be judged on FNA
- **PET/CT** for staging; Lugano / Ann Arbor. Marrow biopsy when it would change management (e.g. apparent limited stage before RT)
- CBC, CMP, **LDH, β2-microglobulin**, **HBsAg + anti-HBc** (reactivation with anti-CD20), HCV, HIV
- **Biopsy the PET-hottest lesion** if SUV is discordant or one node grows out of step — it is the site most likely to have transformed

FLIPI — "NOLASH", 1 POINT EACH

Risk	%	5-yr OS	10-yr OS
Low (0–1)	36	91%	71%
Intermediate (2)	37	78%	51%
High (3–5)	27	53%	**36%**

Nodal areas >4 · **L**DH >ULN · **A**ge ≥60 · **S**tage III–IV · **H**gb <12 (Solal-Céligny, Blood 2004; pre-rituximab era). **FLIPI-2**: age >60, β2M >ULN, marrow involved, Hgb <12, largest node >6 cm — 5-yr PFS 80% / 51% / **19%**. **Neither index tells you when to treat** — GELF does.

TREATMENT PATHWAY

Limited stage (I–contiguous II)

↓

ISRT 24 Gy / 12 fractions
potentially curative; 24 Gy = 40–45 Gy (Lowry)
4 Gy ×2 is inferior for durable control (FoRT)
adding R-CVP to RT improved PFS, not OS (TROG 99.03)

Advanced stage

↓

GELF-negative, asymptomatic
observe · or rituximab 375 weekly ×4

↓

GELF-positive (high burden)
BR (StiL) · R-CHOP · R-CVP
or obinutuzumab with chemo (GALLIUM)
or R² — chemo-free, equivalent (RELEVANCE)
older/unfit: rituximab alone ± chlorambucil

↓

Maintenance (optional)
rituximab q8wk ×2 yr (PRIMA) or obinutuzumab
PFS gain, no OS gain — weigh infections

Relapse

↓

Rebiopsy first
exclude transformation — **mandatory after POD24**

↓

2L+ (any time after 1 prior line)
epcoritamab + R² (EPCORE FL-1) · tafasitamab + R² R² (AUGMENT) · alternative chemoimmunotherapy
chemosensitive POD24: consider auto-HCT

↓

3L+
CAR-T: axi-cel, liso-cel, tisa-cel
bispecifics: mosunetuzumab, epcoritamab
zanubrutinib + obinutuzumab (ROSEWOOD)
allo-HCT: young, fit, multiply relapsed

GELF — ANY ONE = HIGH BURDEN, TREAT

- **≥3 nodal sites, each ≥3 cm**
- Any nodal or extranodal mass **≥7 cm**
- B symptoms or other disease-related symptoms
- Splenomegaly · effusion or ascites · organ compression
- **Cytopenias** (ANC <1,000 or platelets <100,000) · leukemic phase >5,000/μL

Brice, JCO 1997. **Meeting none = observe.** Watch-and-wait has never lost survival against early treatment; rituximab ×4 delays the need for next therapy (3-yr no new treatment **88% vs 46%**, HR 0.21; Ardeshna, Lancet Oncol 2014) but has not improved OS.

FIRST-LINE EVIDENCE

Question	Answer
BR vs R-CHOP *StiL NHL1*	BR: median PFS **69.5 vs 31.2 mo**, HR 0.58, far less toxicity (no alopecia, less neuropathy). **No OS difference**
Obinutuzumab vs rituximab *GALLIUM*	7-yr PFS **63.4% vs 55.7%**; **no OS gain**. More serious AEs (48.9% vs 43.4%) and infusion reactions. **With bendamustine: more grade 3–5 infections, second cancers and fatal events** — prolonged T-cell depletion
R² vs R-chemo *RELEVANCE*	**Equivalent chemo-free option** — 6-yr PFS 60% vs 59%, OS 89% both; less neutropenia, more rash
Maintenance *PRIMA*	Median PFS **10.5 vs 4.1 yr** at 9 yr follow-up; **no OS benefit**; grade 2–4 infections 39% vs 24%
Low burden: retreat or maintain? *RESORT*	Rituximab retreatment at progression = maintenance for time to treatment failure (3.9 vs 4.3 yr), using **4 vs 18 doses**

> **POD24 is the prognostic event that matters.** ~20% progress within 2 years of first-line chemoimmunotherapy; their **5-yr OS is ~50% vs ~90%** (FLIPI-adjusted HR 6.4; Casulo, JCO 2015). Implications: **rebiopsy** (a large share are transformed), avoid simply repeating the same chemo, and move early to **R²-based triplets, CAR-T or bispecifics**; consolidate chemosensitive relapse with auto-HCT in selected patients.

PEARLS

- **FLIPI prognosticates; GELF decides.** A high FLIPI with no GELF criterion is still a reason to observe
- **No first-line upgrade has improved survival** — not obinutuzumab, not maintenance, not R². Choose on toxicity and patient priorities
- **Old grade 3B (FLBL) gets R-CHOP**, not BR; grade 3A is treated as FL
- **Rebiopsy every progression that looks different** — POD24, a discordant node, a hot PET lesion, a rising LDH
- **Think ahead to CAR-T** before giving bendamustine to a high-risk patient

Follicular lymphoma

RELAPSED / REFRACTORY — AGENTS

Agent · US status	Evidence
R² (lenalidomide + rituximab) *Regular approval (FL, MZL; 2019)*	**AUGMENT** (phase 3, FL + MZL, not rituximab-refractory): PFS **39.4 vs 14.1 mo**, HR 0.46; 5-yr update **OS HR 0.59**
Epcoritamab + R² *Approved 2L+, Nov 2025*	**EPCORE FL-1** (phase 3, ≥1 prior line): ORR **95% vs 79%**, PFS **HR 0.21**, 16-mo PFS 85.5% vs 40.2%. Grade ≥3 AEs 90% vs 68%; CRS low grade (gr 2 in 5%). **OS immature**
Tafasitamab + R² *Approved 2L+, Jun 2025*	**inMIND** (phase 3, double-blind): median PFS **22.4 vs 13.9 mo**, HR 0.43. No CRS — a fit for patients unsuited to T-cell engagers
Mosunetuzumab (CD20×CD3, IV) *Accelerated, 3L+*	Single-arm phase 2, 3L+: **CR 60%**; fixed duration — **8 cycles if CR, up to 17**. CRS 44%, mostly grade 1–2. **CELESTIMO** (+ lenalidomide vs R², 2L+) met PFS — topline only, Sept 2026
Epcoritamab (CD20×CD3, SC) *Regular approval, 3L+ (Nov 2025)*	EPCORE NHL-1 FL cohort, 3L+: ORR **82%**, CR **62.5%**; continuous until progression. 3-step-up dosing cut CRS to 49% (none ≥grade 3)
Axi-cel *Accelerated, 3L+*	**ZUMA-5** (single arm, 3L+): ORR 94% FL; median PFS **62.2 mo** (all iNHL) at 5-yr follow-up; few relapses after 30 mo. Grade ≥3 neurotoxicity 19%
Liso-cel *Accelerated, 3L+ (May 2024)*	**TRANSCEND FL**: 3L+ ORR **97%**, CR 94%; 2L high-risk (POD24) ORR 96%. CRS 58% (grade ≥3 1%)
Tisa-cel *Accelerated, 3L+*	**ELARA**: CR 69%, ORR 86%; no grade ≥3 CRS
Zanubrutinib + obinutuzumab *Accelerated, 3L+ (Mar 2024)*	**ROSEWOOD** (randomized **phase 2** vs obinutuzumab): ORR **69% vs 46%**, median PFS 28.0 vs 10.4 mo. Confirmatory MAHOGANY (vs R²) ongoing

No head-to-head data between CAR-T and bispecifics. CAR-T gives the most durable single-shot responses; bispecifics are off-the-shelf and outpatient. **Recent bendamustine blunts CAR-T expansion and outcome** (ZUMA-5) — plan the sequence.

WITHDRAWN AGENTS

Agent	What happened
Tazemetostat (EZH2)	Accelerated approval 2020 (3L+ EZH2-mutant, or no alternative). **Withdrawn from the US market 2026**: in SYMPHONY-1 (+ R²), hematologic second cancers (MDS/AML, B-ALL) in **5.7% vs 0%**, 3 deaths
PI3K inhibitors	**Idelalisib, duvelisib** FL indications withdrawn 2021–22; **umbralisib** withdrawn 2022 (excess deaths); **copanlisib** withdrawn Nov 2023 (CHRONOS-4 negative). Class problem: infections, colitis, pneumonitis, hepatitis, and an OS signal
Ibrutinib	Never approved for FL; confirmatory SELENE (FL/MZL) missed PFS

TRANSFORMATION

- **10-yr cumulative risk ~5–8%** in the rituximab era (5.2% with rituximab vs 8.7% without; Aristotle, Lancet Haematol 2018)
- **Suspect it** with discordant nodal growth, new B symptoms, rising LDH, hypercalcemia, new extranodal disease, or **a PET lesion much hotter than the rest**
- **Biopsy the PET-hottest site** — a random node can miss it. Check **MYC**: with the existing BCL2 rearrangement a MYC break makes it **HGBL with MYC + BCL2**
- **Treat as DLBCL:** anthracycline-naïve → R-CHOP; already had R-CHOP → salvage + auto-HCT or **CAR-T** (axi-cel, liso-cel and tisa-cel labels include DLBCL arising from FL)

HOW TO CHOOSE AT RELAPSE

- **Late relapse, low burden:** rituximab again, or R². **Symptomatic, first relapse:** epcoritamab + R² is now the strongest randomized 2L data; tafasitamab + R² if a T-cell engager is not suitable
- **POD24 or double-refractory** (anti-CD20 + alkylator): favor **CAR-T** (liso-cel has 2L POD24 data) or a bispecific; auto-HCT for chemosensitive early relapse
- **Frail:** rituximab alone, R², or zanubrutinib + obinutuzumab. **Palliation of a symptomatic node:** 2 × 2 Gy

REGIMEN DOSING

Regimen	Dosing
Rituximab	**375 mg/m² weekly ×4**; maintenance 375 **q8 wk ×2 yr** (PRIMA). SC rituximab 1400 mg after a first full IV dose
BR	Bendamustine **90 mg/m² d1+2** + rituximab 375 d1, **q28d ×6**
R-CHOP / R-CVP	Rituximab 375, cyclophosphamide 750, vincristine 1.4 (max 2 mg) d1, prednisone 40 mg/m² d1–5, **± doxorubicin 50** — q21d ×6 (CHOP) or ×8 (CVP)
Obinutuzumab	**1000 mg flat, C1 d1, 8, 15**, then d1 of each cycle (C2–6 with benda, C2–8 with CHOP/CVP); maintenance **1000 mg q2 mo ×2 yr**
R²	Lenalidomide **20 mg PO d1–21 q28d ×12** (10 mg if CrCl 30–59); rituximab 375 weekly C1 then d1 C2–5
Epcoritamab + R²	Epcoritamab 48 mg SC after step-up, **weekly C1–3, then q4wk C4–12** (fixed duration) + R² as above
Epcoritamab mono	SC **0.16 → 0.8 → 3 → 48 mg** (C1 d1, 8, 15, 22), 48 mg weekly C2–3, q2wk C4–9, then q4wk until progression
Mosunetuzumab	IV **1 → 2 → 60 mg** (C1 d1, 8, 15), 60 mg C2 d1, then 30 mg q21d; stop after **8 cycles if CR**, else up to 17
Tafasitamab + R²	Tafasitamab **12 mg/kg** d1, 8, 15, 22 C1–3, then d1+15 C4–12 (q28d) + R²
Zanubrutinib + obinutuzumab	Zanubrutinib **160 mg BID** continuous + obinutuzumab 1000 mg (C1 d1, 8, 15; C2–6 d1; then q8wk)
ISRT	**24 Gy / 12 fx** curative intent · **2 × 2 Gy** palliative

mg/m² unless stated. Premedicate every anti-CD20 infusion (acetaminophen, antihistamine, steroid for the first).

TOXICITY AND MONITORING

- **HBV:** screen HBsAg + anti-HBc before any anti-CD20; **entecavir prophylaxis** if either positive, continued for ≥12 months after the last dose
- **Bendamustine → profound, prolonged CD4 lymphopenia** (months to years). Give **PJP prophylaxis (TMP-SMX) and VZV prophylaxis (acyclovir/valacyclovir)**; watch for CMV. Worse when followed by maintenance anti-CD20 — think twice about stacking them. Also hampers later CAR-T manufacturing
- **Obinutuzumab:** infusion reactions ~60% (mostly C1 d1) — premedicate, slow first infusion, hold antihypertensives. More neutropenia and infection than rituximab
- **Bispecifics:** CRS clusters around the **first full dose** of step-up. Corticosteroid premedication, hydration, close monitoring after the first full dose; tocilizumab for grade ≥2. ICANS uncommon. Continuous B-cell depletion → **hypogammaglobulinemia and serious infections**: IVIG if recurrent, PJP + antiviral prophylaxis, vaccinate beforehand
- **R²: VTE** — aspirin 81 mg if no other risk, LMWH/DOAC if high risk; **rash** (often grade 1–2; antihistamine, hold/reduce), neutropenia (50% grade 3–4 in AUGMENT), tumor flare, diarrhea. Dose by renal function
- **CAR-T:** CRS and ICANS (REMS removed June 2025: stay near the center and no driving for **2 weeks**), prolonged cytopenias, hypogammaglobulinemia; T-cell malignancy boxed warning
- **Zanubrutinib:** bleeding (hold 3–7 days peri-procedure), AF, hypertension, neutropenia

KEY TRIALS

StiL NHL1	BR > R-CHOP for PFS; no OS difference
GALLIUM	Obinutuzumab: PFS gain, **no OS gain**
RELEVANCE	R² = R-chemo (chemo-free option)
PRIMA	Maintenance: PFS doubled, **no OS**
FoRT	24 Gy beats 4 Gy for local control
AUGMENT	R² vs R at relapse: PFS HR 0.46
EPCORE FL-1	Epcoritamab + R²: PFS HR 0.21
inMIND	Tafasitamab + R²: PFS HR 0.43